Debit Risk Planner an Agent Studio agent that stops subscription renewals failing before the debit

Track 2: Recover and Grow with AI · Saket, ISB PGP · Prototype: link in submission · All results on synthetic data

Problem

About **20% of repeat UPI AutoPay debits fail** (Razorpay, Mar 2026); roughly two-thirds are low balance. Every Razorpay recovery agent acts **after** the failure, by which time the customer has lost access and is one tap from cancelling the mandate in their UPI app. SME subscription founders (edtech, fitness, D2C) have no billing team; they see the damage only as churn.

Insight

Most failures are **timing, not inability to pay**. Priya pays ₹2,999 to SkillUp on the 1st but is paid on the 7th; her four failed renewals were all paid days later. Today's fixed retries (next three days) use up NPCI's 1 + 3 attempts before her salary arrives, so the subscription halts.

Solution

Each night the agent scores every renewal due in the next 7 days, picks the cheapest action worth taking, and sends the merchant one weekly plan to approve on RAY (WhatsApp) or the dashboard. It executes through existing Razorpay rails and hands any remaining failure to the existing recovery agents with a diagnosis. **The model predicts, the rules protect, the agent decides and explains, the merchant approves.**

Track 2 question	Our answer
Signal	Upcoming debits + failure history and reason codes, likely payday (from late-payment timing and Razorpay inflow signals), bank and rail, amount vs mandate limit, card expiry, scheduled bank downtime
Why AI, not a rule	Risk is a non-linear mix of these signals per customer; reminder targeting is an expected-value trade-off (₹ saved vs cost and annoyance). "Remind everyone" lost money in testing. The LLM agent explains, drafts in the customer's language, turns typed instructions into rules
Action	Ladder: nothing → retry timed after payday → re-approve limit / update card → WhatsApp reminder 2 days early → hand to recovery agent. Holds messages during bank downtime
Merchant control	Review-first weekly plan, per-action switches, message cap per customer, plain-language rule changes priced in ₹ before saving, one-tap stop
Learning	10% random control group every week; recalibrate risk weekly; update reminder effect and fatigue from the control; champion vs challenger on timing
Trust and failure cases	Contact only with merchant enablement and customer opt-in; opt-outs permanent; no discounts or pressure; fall back to standard retries if unsure or model down; NPCI windows and attempt limits enforced in code
Outcome	North star: ₹ revenue protected vs control, net of message cost. Also first-attempt success, never-collected renewals. Guardrails: cancellations, opt-outs

Evidence (synthetic simulation, 40k subscribers, 5 merchants, out-of-time test, 5 runs)

Policy (2 months)	Net value vs today	Cancellations	Messages
Remind everyone	−₹5.6 L	+7%	60.7k
Best fixed rules (smart retries, mandate fixes, remind if failed last month)	₹62.7 L	−16%	6.9k
Debit Risk Planner	**₹65.6 L**	**−16%**	25.3k

SkillUp: collection 92.4% → 95.2%, about **₹5.8 L more cash a month**. Risk model AUC 0.74, calibrated by decile. **Sensitivity:** if reminders work worse than assumed, a planner with fixed assumptions loses to good rules; once it learns from the control group it beats them in every scenario tested (it simply stops sending reminders that don't work).

Decisions and what I deliberately did not build

- **No customer "postpone" button**: it gives free access and can be gamed. A one-time, prorated billing-date move exists but is **off by default**; it added no value in simulation.
- **No chatbot dashboard, no lending EMIs** (crowded, regulated), **no discounts** by the agent, no other merchants' data ever shown to a merchant.
- Most of the gain comes from acting before the debit with rules; AI's edge (~5%) depends on learning. I built the learning loop in from day one rather than overclaiming the model.

Assumptions and validation

Synthetic data calibrated to public sources: Razorpay failure rates, NPCI bank-wise declines, PLFS 2023-24 income mix, reminder trials (Uganda, Philippines). Unverified: reminder fatigue, churn after failure, payday signal accuracy, retry timing allowed up to ~10 days. **Pilot:** 3 merchants, 4–8 weeks, 10% control, reminders to the riskiest decile first.